

Newman-Penrose Formalism Calculations

Stephani et al., Equation (kerr)

Metric

$$ds^2 = \left(-\frac{a^2 \cos(\theta)^2 + Q^2 - 2Mr + r^2}{a^2 \cos(\theta)^2 + r^2} \right) dt \otimes dt + \left(\frac{(Q^2 a - 2Mar) \sin(\theta)^2}{a^2 \cos(\theta)^2 + r^2} \right) dt \otimes d\phi + \left(\frac{a^2 \cos(\theta)^2 + r^2}{Q^2 + a^2 - 2Mr + r^2} \right) dr \otimes dr + (a^2 \cos(\theta)^2 + r^2) d\theta \otimes d\theta + \left(\frac{(Q^2 a - 2Mar) \sin(\theta)^2}{a^2 \cos(\theta)^2 + r^2} \right) d\phi \otimes dt + \left(-\frac{(Q^2 a^2 + a^4 - 2Ma^2 r + a^2 r^2) \sin(\theta)^4 - (a^4 + 2a^2 r^2 + r^4) \sin(\theta)^2}{a^2 \cos(\theta)^2 + r^2} \right) d\phi \otimes d\phi$$

Coordinates

$$(x^\mu) = (t, r, \theta, \phi)$$

Null Tetrad

Covariant components

$$l = (-1) dt + \left(\frac{a^2 \cos(\theta)^2}{Q^2 + a^2 - 2Mr + r^2} + \frac{r^2}{Q^2 + a^2 - 2Mr + r^2} \right) dr + (a \sin(\theta)^2) d\phi$$

$$n = \left(-\frac{Q^2}{2(a^2 \cos(\theta)^2 + r^2)} - \frac{a^2}{2(a^2 \cos(\theta)^2 + r^2)} + \frac{Mr}{a^2 \cos(\theta)^2 + r^2} - \frac{r^2}{2(a^2 \cos(\theta)^2 + r^2)} \right) dt + \left(-\frac{1}{2} \right) dr + \left(\frac{Q^2 a \sin(\theta)^2}{2(a^2 \cos(\theta)^2 + r^2)} + \frac{a^3 \sin(\theta)^2}{2(a^2 \cos(\theta)^2 + r^2)} - \frac{Mar \sin(\theta)^2}{a^2 \cos(\theta)^2 + r^2} + \frac{ar^2 \sin(\theta)^2}{2(a^2 \cos(\theta)^2 + r^2)} \right) d\phi$$

$$m = \left(-\frac{\sqrt{2}a^2 \cos(\theta) \sin(\theta)}{2(a^2 \cos(\theta)^2 + r^2)} - \frac{i\sqrt{2}ar \sin(\theta)}{2(a^2 \cos(\theta)^2 + r^2)} \right) dt + \left(-\frac{1}{2}i\sqrt{2}a \cos(\theta) + \frac{1}{2}\sqrt{2}r \right) d\theta + \left(\frac{\sqrt{2}a^3 \cos(\theta) \sin(\theta)}{2(a^2 \cos(\theta)^2 + r^2)} + \frac{\sqrt{2}ar^2 \cos(\theta) \sin(\theta)}{2(a^2 \cos(\theta)^2 + r^2)} + \frac{i\sqrt{2}a^2 r \sin(\theta)}{2(a^2 \cos(\theta)^2 + r^2)} + \frac{i\sqrt{2}r^3 \sin(\theta)}{2(a^2 \cos(\theta)^2 + r^2)} \right) d\phi$$

$$\begin{aligned}\bar{m} = & \left(-\frac{\sqrt{2}a^2 \cos(\theta) \sin(\theta)}{2(a^2 \cos(\theta)^2 + r^2)} + \frac{i\sqrt{2}ar \sin(\theta)}{2(a^2 \cos(\theta)^2 + r^2)} \right) dt + \left(\frac{1}{2}i\sqrt{2}a \cos(\theta) + \frac{1}{2}\sqrt{2}r \right) d\theta \\ & + \left(\frac{\sqrt{2}a^3 \cos(\theta) \sin(\theta)}{2(a^2 \cos(\theta)^2 + r^2)} + \frac{\sqrt{2}ar^2 \cos(\theta) \sin(\theta)}{2(a^2 \cos(\theta)^2 + r^2)} - \frac{i\sqrt{2}a^2r \sin(\theta)}{2(a^2 \cos(\theta)^2 + r^2)} - \frac{i\sqrt{2}r^3 \sin(\theta)}{2(a^2 \cos(\theta)^2 + r^2)} \right) d\phi\end{aligned}$$

Contravariant components

$$l = \left(\frac{a^2}{Q^2 + a^2 - 2Mr + r^2} + \frac{r^2}{Q^2 + a^2 - 2Mr + r^2} \right) dt + (1) dr + \left(\frac{a}{Q^2 + a^2 - 2Mr + r^2} \right) d\phi$$

$$\begin{aligned}n = & \left(\frac{a^2}{2(a^2 \cos(\theta)^2 + r^2)} + \frac{r^2}{2(a^2 \cos(\theta)^2 + r^2)} \right) dt \\ & + \left(-\frac{Q^2}{2(a^2 \cos(\theta)^2 + r^2)} - \frac{a^2}{2(a^2 \cos(\theta)^2 + r^2)} + \frac{Mr}{a^2 \cos(\theta)^2 + r^2} - \frac{r^2}{2(a^2 \cos(\theta)^2 + r^2)} \right) dr + \left(\frac{a}{2(a^2 \cos(\theta)^2 + r^2)} \right) d\phi\end{aligned}$$

$$m = \left(\frac{i\sqrt{2}a \sin(\theta)}{2(i a \cos(\theta) + r)} \right) dt + \left(\frac{\sqrt{2}}{2(i a \cos(\theta) + r)} \right) d\theta + \left(\frac{i\sqrt{2}}{2(i a \cos(\theta) + r) \sin(\theta)} \right) d\phi$$

$$\bar{m} = \left(-\frac{i\sqrt{2}a \sin(\theta)}{2(-i a \cos(\theta) + r)} \right) dt + \left(\frac{\sqrt{2}}{2(-i a \cos(\theta) + r)} \right) d\theta + \left(-\frac{i\sqrt{2}}{2(-i a \cos(\theta) + r) \sin(\theta)} \right) d\phi$$

Directional Derivatives

$$D = \frac{a^2}{Q^2 + a^2 - 2Mr + r^2} + \frac{r^2}{Q^2 + a^2 - 2Mr + r^2} \partial_t + \partial_r + \frac{a}{Q^2 + a^2 - 2Mr + r^2} \partial_\phi$$

$$\Delta = \frac{a^2}{2(a^2 \cos(\theta)^2 + r^2)} + \frac{r^2}{2(a^2 \cos(\theta)^2 + r^2)} \partial_t - \frac{Q^2}{2(a^2 \cos(\theta)^2 + r^2)} - \frac{a^2}{2(a^2 \cos(\theta)^2 + r^2)} + \frac{Mr}{a^2 \cos(\theta)^2 + r^2} - \frac{r^2}{2(a^2 \cos(\theta)^2 + r^2)} \partial_r + \frac{a}{2(a^2 \cos(\theta)^2 + r^2)} \partial_\phi$$

$$\delta = \frac{i\sqrt{2}a\sin(\theta)}{2(i a \cos(\theta) + r)} \partial_t + \frac{\sqrt{2}}{2(i a \cos(\theta) + r)} \partial_\theta + \frac{i\sqrt{2}}{2(i a \cos(\theta) + r)\sin(\theta)} \partial_\phi$$

$$\bar{\delta} = -\frac{i\sqrt{2}a\sin(\theta)}{2(-i a \cos(\theta) + r)} \partial_t + \frac{\sqrt{2}}{2(-i a \cos(\theta) + r)} \partial_\theta - \frac{i\sqrt{2}}{2(-i a \cos(\theta) + r)\sin(\theta)} \partial_\phi$$

Spin Coefficients

$$\rho = -\frac{i a \cos(\theta)}{a^2 \cos(\theta)^2 + r^2} - \frac{r}{a^2 \cos(\theta)^2 + r^2}$$

$$\sigma = 0$$

$$\kappa = 0$$

$$\begin{aligned} \mu = & -\frac{i Q^2 a \cos(\theta)}{2 \left(a^4 \cos(\theta)^4 + 2 a^2 r^2 \cos(\theta)^2 + r^4 \right)} - \frac{i a^3 \cos(\theta)}{2 \left(a^4 \cos(\theta)^4 + 2 a^2 r^2 \cos(\theta)^2 + r^4 \right)} + \frac{i M a r \cos(\theta)}{a^4 \cos(\theta)^4 + 2 a^2 r^2 \cos(\theta)^2 + r^4} - \frac{i a r^2 \cos(\theta)}{2 \left(a^4 \cos(\theta)^4 + 2 a^2 r^2 \cos(\theta)^2 + r^4 \right)} \\ & - \frac{Q^2 r}{2 \left(a^4 \cos(\theta)^4 + 2 a^2 r^2 \cos(\theta)^2 + r^4 \right)} - \frac{a^2 r}{2 \left(a^4 \cos(\theta)^4 + 2 a^2 r^2 \cos(\theta)^2 + r^4 \right)} + \frac{M r^2}{a^4 \cos(\theta)^4 + 2 a^2 r^2 \cos(\theta)^2 + r^4} - \frac{r^3}{2 \left(a^4 \cos(\theta)^4 + 2 a^2 r^2 \cos(\theta)^2 + r^4 \right)} \end{aligned}$$

$$\lambda = 0$$

$$\nu = 0$$

$\alpha =$

$$\begin{aligned}
& - \frac{a^8 \cos(\theta)^9}{2 \left(i \sqrt{2} a^9 \cos(\theta)^9 - 9 \sqrt{2} a^8 r \cos(\theta)^8 - 36 i \sqrt{2} a^7 r^2 \cos(\theta)^7 + 84 \sqrt{2} a^6 r^3 \cos(\theta)^6 + 126 i \sqrt{2} a^5 r^4 \cos(\theta)^5 - 126 \sqrt{2} a^4 r^5 \cos(\theta)^4 - 84 i \sqrt{2} a^3 r^6 \cos(\theta)^3 + 36 \sqrt{2} a^2 r^7 \cos(\theta)^2 + 9 i \sqrt{2} a r^8 \cos(\theta) - r^9 \right)} \\
& - \frac{3 i a^7 r \cos(\theta)^8}{\left(i \sqrt{2} a^9 \cos(\theta)^9 - 9 \sqrt{2} a^8 r \cos(\theta)^8 - 36 i \sqrt{2} a^7 r^2 \cos(\theta)^7 + 84 \sqrt{2} a^6 r^3 \cos(\theta)^6 + 126 i \sqrt{2} a^5 r^4 \cos(\theta)^5 - 126 \sqrt{2} a^4 r^5 \cos(\theta)^4 - 84 i \sqrt{2} a^3 r^6 \cos(\theta)^3 + 36 \sqrt{2} a^2 r^7 \cos(\theta)^2 + 9 i \sqrt{2} a r^8 \cos(\theta) - r^9 \right)} \\
& + \frac{a^8 \cos(\theta)^7}{\left(i \sqrt{2} a^9 \cos(\theta)^9 - 9 \sqrt{2} a^8 r \cos(\theta)^8 - 36 i \sqrt{2} a^7 r^2 \cos(\theta)^7 + 84 \sqrt{2} a^6 r^3 \cos(\theta)^6 + 126 i \sqrt{2} a^5 r^4 \cos(\theta)^5 - 126 \sqrt{2} a^4 r^5 \cos(\theta)^4 - 84 i \sqrt{2} a^3 r^6 \cos(\theta)^3 + 36 \sqrt{2} a^2 r^7 \cos(\theta)^2 + 9 i \sqrt{2} a r^8 \cos(\theta) - r^9 \right)} \\
& + \frac{7 a^6 r^2 \cos(\theta)^7}{\left(i \sqrt{2} a^9 \cos(\theta)^9 - 9 \sqrt{2} a^8 r \cos(\theta)^8 - 36 i \sqrt{2} a^7 r^2 \cos(\theta)^7 + 84 \sqrt{2} a^6 r^3 \cos(\theta)^6 + 126 i \sqrt{2} a^5 r^4 \cos(\theta)^5 - 126 \sqrt{2} a^4 r^5 \cos(\theta)^4 - 84 i \sqrt{2} a^3 r^6 \cos(\theta)^3 + 36 \sqrt{2} a^2 r^7 \cos(\theta)^2 + 9 i \sqrt{2} a r^8 \cos(\theta) - r^9 \right)} \\
& + \frac{7 i a^7 r \cos(\theta)^6}{\left(i \sqrt{2} a^9 \cos(\theta)^9 - 9 \sqrt{2} a^8 r \cos(\theta)^8 - 36 i \sqrt{2} a^7 r^2 \cos(\theta)^7 + 84 \sqrt{2} a^6 r^3 \cos(\theta)^6 + 126 i \sqrt{2} a^5 r^4 \cos(\theta)^5 - 126 \sqrt{2} a^4 r^5 \cos(\theta)^4 - 84 i \sqrt{2} a^3 r^6 \cos(\theta)^3 + 36 \sqrt{2} a^2 r^7 \cos(\theta)^2 + 9 i \sqrt{2} a r^8 \cos(\theta) - r^9 \right)} \\
& + \frac{7 i a^5 r^3 \cos(\theta)^6}{\left(i \sqrt{2} a^9 \cos(\theta)^9 - 9 \sqrt{2} a^8 r \cos(\theta)^8 - 36 i \sqrt{2} a^7 r^2 \cos(\theta)^7 + 84 \sqrt{2} a^6 r^3 \cos(\theta)^6 + 126 i \sqrt{2} a^5 r^4 \cos(\theta)^5 - 126 \sqrt{2} a^4 r^5 \cos(\theta)^4 - 84 i \sqrt{2} a^3 r^6 \cos(\theta)^3 + 36 \sqrt{2} a^2 r^7 \cos(\theta)^2 + 9 i \sqrt{2} a r^8 \cos(\theta) - r^9 \right)} \\
& - \frac{21 a^6 r^2 \cos(\theta)^5}{\left(i \sqrt{2} a^9 \cos(\theta)^9 - 9 \sqrt{2} a^8 r \cos(\theta)^8 - 36 i \sqrt{2} a^7 r^2 \cos(\theta)^7 + 84 \sqrt{2} a^6 r^3 \cos(\theta)^6 + 126 i \sqrt{2} a^5 r^4 \cos(\theta)^5 - 126 \sqrt{2} a^4 r^5 \cos(\theta)^4 - 84 i \sqrt{2} a^3 r^6 \cos(\theta)^3 + 36 \sqrt{2} a^2 r^7 \cos(\theta)^2 + 9 i \sqrt{2} a r^8 \cos(\theta) - r^9 \right)} \\
& - \dots
\end{aligned}$$

$$\begin{aligned}
\beta = & - \frac{i \sqrt{2} a^5 \cos(\theta)^6}{4 \left(a^6 \cos(\theta)^6 - 6 i a^5 r \cos(\theta)^5 - 15 a^4 r^2 \cos(\theta)^4 + 20 i a^3 r^3 \cos(\theta)^3 + 15 a^2 r^4 \cos(\theta)^2 - 6 i a r^5 \cos(\theta) - r^6 \right) \sin(\theta)} \\
& - \frac{5 \sqrt{2} a^4 r \cos(\theta)^5}{4 \left(a^6 \cos(\theta)^6 - 6 i a^5 r \cos(\theta)^5 - 15 a^4 r^2 \cos(\theta)^4 + 20 i a^3 r^3 \cos(\theta)^3 + 15 a^2 r^4 \cos(\theta)^2 - 6 i a r^5 \cos(\theta) - r^6 \right) \sin(\theta)} \\
& + \frac{5 i \sqrt{2} a^3 r^2 \cos(\theta)^4}{2 \left(a^6 \cos(\theta)^6 - 6 i a^5 r \cos(\theta)^5 - 15 a^4 r^2 \cos(\theta)^4 + 20 i a^3 r^3 \cos(\theta)^3 + 15 a^2 r^4 \cos(\theta)^2 - 6 i a r^5 \cos(\theta) - r^6 \right) \sin(\theta)} \\
& + \frac{5 \sqrt{2} a^2 r^3 \cos(\theta)^3}{2 \left(a^6 \cos(\theta)^6 - 6 i a^5 r \cos(\theta)^5 - 15 a^4 r^2 \cos(\theta)^4 + 20 i a^3 r^3 \cos(\theta)^3 + 15 a^2 r^4 \cos(\theta)^2 - 6 i a r^5 \cos(\theta) - r^6 \right) \sin(\theta)} \\
& - \frac{5 i \sqrt{2} a r^4 \cos(\theta)^2}{4 \left(a^6 \cos(\theta)^6 - 6 i a^5 r \cos(\theta)^5 - 15 a^4 r^2 \cos(\theta)^4 + 20 i a^3 r^3 \cos(\theta)^3 + 15 a^2 r^4 \cos(\theta)^2 - 6 i a r^5 \cos(\theta) - r^6 \right) \sin(\theta)} \\
& - \frac{\sqrt{2} r^5 \cos(\theta)}{4 \left(a^6 \cos(\theta)^6 - 6 i a^5 r \cos(\theta)^5 - 15 a^4 r^2 \cos(\theta)^4 + 20 i a^3 r^3 \cos(\theta)^3 + 15 a^2 r^4 \cos(\theta)^2 - 6 i a r^5 \cos(\theta) - r^6 \right) \sin(\theta)}
\end{aligned}$$

$$\begin{aligned}
\tau = & - \frac{i \sqrt{2} a^9 \cos(\theta)^8 \sin(\theta)}{2 \left(a^{10} \cos(\theta)^{10} - 8i a^9 r \cos(\theta)^9 - 27 a^8 r^2 \cos(\theta)^8 + 48i a^7 r^3 \cos(\theta)^7 + 42 a^6 r^4 \cos(\theta)^6 + 42 a^4 r^6 \cos(\theta)^4 - 48i a^3 r^7 \cos(\theta)^3 - 27 a^2 r^8 \cos(\theta)^2 + 8i a r^9 \cos(\theta) + r^{10} \right)} \\
& - \frac{4 \sqrt{2} a^8 r \cos(\theta)^7 \sin(\theta)}{a^{10} \cos(\theta)^{10} - 8i a^9 r \cos(\theta)^9 - 27 a^8 r^2 \cos(\theta)^8 + 48i a^7 r^3 \cos(\theta)^7 + 42 a^6 r^4 \cos(\theta)^6 + 42 a^4 r^6 \cos(\theta)^4 - 48i a^3 r^7 \cos(\theta)^3 - 27 a^2 r^8 \cos(\theta)^2 + 8i a r^9 \cos(\theta) + r^{10}} \\
& + \frac{14i \sqrt{2} a^7 r^2 \cos(\theta)^6 \sin(\theta)}{a^{10} \cos(\theta)^{10} - 8i a^9 r \cos(\theta)^9 - 27 a^8 r^2 \cos(\theta)^8 + 48i a^7 r^3 \cos(\theta)^7 + 42 a^6 r^4 \cos(\theta)^6 + 42 a^4 r^6 \cos(\theta)^4 - 48i a^3 r^7 \cos(\theta)^3 - 27 a^2 r^8 \cos(\theta)^2 + 8i a r^9 \cos(\theta) + r^{10}} \\
& + \frac{28 \sqrt{2} a^6 r^3 \cos(\theta)^5 \sin(\theta)}{a^{10} \cos(\theta)^{10} - 8i a^9 r \cos(\theta)^9 - 27 a^8 r^2 \cos(\theta)^8 + 48i a^7 r^3 \cos(\theta)^7 + 42 a^6 r^4 \cos(\theta)^6 + 42 a^4 r^6 \cos(\theta)^4 - 48i a^3 r^7 \cos(\theta)^3 - 27 a^2 r^8 \cos(\theta)^2 + 8i a r^9 \cos(\theta) + r^{10}} \\
& - \frac{35i \sqrt{2} a^5 r^4 \cos(\theta)^4 \sin(\theta)}{a^{10} \cos(\theta)^{10} - 8i a^9 r \cos(\theta)^9 - 27 a^8 r^2 \cos(\theta)^8 + 48i a^7 r^3 \cos(\theta)^7 + 42 a^6 r^4 \cos(\theta)^6 + 42 a^4 r^6 \cos(\theta)^4 - 48i a^3 r^7 \cos(\theta)^3 - 27 a^2 r^8 \cos(\theta)^2 + 8i a r^9 \cos(\theta) + r^{10}} \\
& - \frac{28 \sqrt{2} a^4 r^5 \cos(\theta)^3 \sin(\theta)}{a^{10} \cos(\theta)^{10} - 8i a^9 r \cos(\theta)^9 - 27 a^8 r^2 \cos(\theta)^8 + 48i a^7 r^3 \cos(\theta)^7 + 42 a^6 r^4 \cos(\theta)^6 + 42 a^4 r^6 \cos(\theta)^4 - 48i a^3 r^7 \cos(\theta)^3 - 27 a^2 r^8 \cos(\theta)^2 + 8i a r^9 \cos(\theta) + r^{10}} \\
& + \frac{14i \sqrt{2} a^3 r^6 \cos(\theta)^2 \sin(\theta)}{a^{10} \cos(\theta)^{10} - 8i a^9 r \cos(\theta)^9 - 27 a^8 r^2 \cos(\theta)^8 + 48i a^7 r^3 \cos(\theta)^7 + 42 a^6 r^4 \cos(\theta)^6 + 42 a^4 r^6 \cos(\theta)^4 - 48i a^3 r^7 \cos(\theta)^3 - 27 a^2 r^8 \cos(\theta)^2 + 8i a r^9 \cos(\theta) + r^{10}} \\
& + \frac{4 \sqrt{2} a^2 r^7 \cos(\theta) \sin(\theta)}{a^{10} \cos(\theta)^{10} - 8i a^9 r \cos(\theta)^9 - 27 a^8 r^2 \cos(\theta)^8 + 48i a^7 r^3 \cos(\theta)^7 + 42 a^6 r^4 \cos(\theta)^6 + 42 a^4 r^6 \cos(\theta)^4 - 48i a^3 r^7 \cos(\theta)^3 - 27 a^2 r^8 \cos(\theta)^2 + 8i a r^9 \cos(\theta) + r^{10}} \\
& - \frac{i \sqrt{2} a r^8 \sin(\theta)}{2 \left(a^{10} \cos(\theta)^{10} - 8i a^9 r \cos(\theta)^9 - 27 a^8 r^2 \cos(\theta)^8 + 48i a^7 r^3 \cos(\theta)^7 + 42 a^6 r^4 \cos(\theta)^6 + 42 a^4 r^6 \cos(\theta)^4 - 48i a^3 r^7 \cos(\theta)^3 - 27 a^2 r^8 \cos(\theta)^2 + 8i a r^9 \cos(\theta) + r^{10} \right)}
\end{aligned}$$

$$\begin{aligned}
\pi = & - \frac{i \sqrt{2} a^7 \cos(\theta)^6 \sin(\theta)}{2 \left(a^8 \cos(\theta)^8 + 8i a^7 r \cos(\theta)^7 - 28 a^6 r^2 \cos(\theta)^6 - 56i a^5 r^3 \cos(\theta)^5 + 70 a^4 r^4 \cos(\theta)^4 + 56i a^3 r^5 \cos(\theta)^3 - 28 a^2 r^6 \cos(\theta)^2 - 8i a r^7 \cos(\theta) + r^8 \right)} \\
& + \frac{3 \sqrt{2} a^6 r \cos(\theta)^5 \sin(\theta)}{a^8 \cos(\theta)^8 + 8i a^7 r \cos(\theta)^7 - 28 a^6 r^2 \cos(\theta)^6 - 56i a^5 r^3 \cos(\theta)^5 + 70 a^4 r^4 \cos(\theta)^4 + 56i a^3 r^5 \cos(\theta)^3 - 28 a^2 r^6 \cos(\theta)^2 - 8i a r^7 \cos(\theta) + r^8} \\
& + \frac{15i \sqrt{2} a^5 r^2 \cos(\theta)^4 \sin(\theta)}{2 \left(a^8 \cos(\theta)^8 + 8i a^7 r \cos(\theta)^7 - 28 a^6 r^2 \cos(\theta)^6 - 56i a^5 r^3 \cos(\theta)^5 + 70 a^4 r^4 \cos(\theta)^4 + 56i a^3 r^5 \cos(\theta)^3 - 28 a^2 r^6 \cos(\theta)^2 - 8i a r^7 \cos(\theta) + r^8 \right)} \\
& - \frac{10 \sqrt{2} a^4 r^3 \cos(\theta)^3 \sin(\theta)}{a^8 \cos(\theta)^8 + 8i a^7 r \cos(\theta)^7 - 28 a^6 r^2 \cos(\theta)^6 - 56i a^5 r^3 \cos(\theta)^5 + 70 a^4 r^4 \cos(\theta)^4 + 56i a^3 r^5 \cos(\theta)^3 - 28 a^2 r^6 \cos(\theta)^2 - 8i a r^7 \cos(\theta) + r^8} \\
& - \frac{15i \sqrt{2} a^3 r^4 \cos(\theta)^2 \sin(\theta)}{2 \left(a^8 \cos(\theta)^8 + 8i a^7 r \cos(\theta)^7 - 28 a^6 r^2 \cos(\theta)^6 - 56i a^5 r^3 \cos(\theta)^5 + 70 a^4 r^4 \cos(\theta)^4 + 56i a^3 r^5 \cos(\theta)^3 - 28 a^2 r^6 \cos(\theta)^2 - 8i a r^7 \cos(\theta) + r^8 \right)} \\
& + \frac{3 \sqrt{2} a^2 r^5 \cos(\theta) \sin(\theta)}{a^8 \cos(\theta)^8 + 8i a^7 r \cos(\theta)^7 - 28 a^6 r^2 \cos(\theta)^6 - 56i a^5 r^3 \cos(\theta)^5 + 70 a^4 r^4 \cos(\theta)^4 + 56i a^3 r^5 \cos(\theta)^3 - 28 a^2 r^6 \cos(\theta)^2 - 8i a r^7 \cos(\theta) + r^8} \\
& + \frac{i \sqrt{2} a r^6 \sin(\theta)}{2 \left(a^8 \cos(\theta)^8 + 8i a^7 r \cos(\theta)^7 - 28 a^6 r^2 \cos(\theta)^6 - 56i a^5 r^3 \cos(\theta)^5 + 70 a^4 r^4 \cos(\theta)^4 + 56i a^3 r^5 \cos(\theta)^3 - 28 a^2 r^6 \cos(\theta)^2 - 8i a r^7 \cos(\theta) + r^8 \right)}
\end{aligned}$$

$$\epsilon = 0$$

$$\begin{aligned}
\gamma = & - \frac{i M a^{11} \cos(\theta)^{11}}{2 \left(i a^{13} \cos(\theta)^{13} + 9 a^{12} r \cos(\theta)^{12} - 34i a^{11} r^2 \cos(\theta)^{11} - 66 a^{10} r^3 \cos(\theta)^{10} + 55i a^9 r^4 \cos(\theta)^9 - 33 a^8 r^5 \cos(\theta)^8 + 132i a^7 r^6 \cos(\theta)^7 + 132 a^6 r^7 \cos(\theta)^6 - 33i a^5 r^8 \cos(\theta)^5 + 55 a^4 r^9 \cos(\theta)^4 - 33 a^3 r^{10} \cos(\theta)^3 + 5 a^2 r^{11} \cos(\theta)^2 - a r^{12} \cos(\theta) + r^{13} \right)} \\
& + \frac{i a^{11} r \cos(\theta)^{11}}{2 \left(i a^{13} \cos(\theta)^{13} + 9 a^{12} r \cos(\theta)^{12} - 34i a^{11} r^2 \cos(\theta)^{11} - 66 a^{10} r^3 \cos(\theta)^{10} + 55i a^9 r^4 \cos(\theta)^9 - 33 a^8 r^5 \cos(\theta)^8 + 132i a^7 r^6 \cos(\theta)^7 + 132 a^6 r^7 \cos(\theta)^6 - 33i a^5 r^8 \cos(\theta)^5 + 55 a^4 r^9 \cos(\theta)^4 - 33 a^3 r^{10} \cos(\theta)^3 + 5 a^2 r^{11} \cos(\theta)^2 - a r^{12} \cos(\theta) + r^{13} \right)} \\
& + \frac{Q^2 a^{10} \cos(\theta)^{10}}{2 \left(i a^{13} \cos(\theta)^{13} + 9 a^{12} r \cos(\theta)^{12} - 34i a^{11} r^2 \cos(\theta)^{11} - 66 a^{10} r^3 \cos(\theta)^{10} + 55i a^9 r^4 \cos(\theta)^9 - 33 a^8 r^5 \cos(\theta)^8 + 132i a^7 r^6 \cos(\theta)^7 + 132 a^6 r^7 \cos(\theta)^6 - 33i a^5 r^8 \cos(\theta)^5 + 55 a^4 r^9 \cos(\theta)^4 - 33 a^3 r^{10} \cos(\theta)^3 + 5 a^2 r^{11} \cos(\theta)^2 - a r^{12} \cos(\theta) + r^{13} \right)} \\
& + \frac{a^{12} \cos(\theta)^{10}}{2 \left(i a^{13} \cos(\theta)^{13} + 9 a^{12} r \cos(\theta)^{12} - 34i a^{11} r^2 \cos(\theta)^{11} - 66 a^{10} r^3 \cos(\theta)^{10} + 55i a^9 r^4 \cos(\theta)^9 - 33 a^8 r^5 \cos(\theta)^8 + 132i a^7 r^6 \cos(\theta)^7 + 132 a^6 r^7 \cos(\theta)^6 - 33i a^5 r^8 \cos(\theta)^5 + 55 a^4 r^9 \cos(\theta)^4 - 33 a^3 r^{10} \cos(\theta)^3 + 5 a^2 r^{11} \cos(\theta)^2 - a r^{12} \cos(\theta) + r^{13} \right)} \\
& - \frac{11 M a^{10} r \cos(\theta)^{10}}{2 \left(i a^{13} \cos(\theta)^{13} + 9 a^{12} r \cos(\theta)^{12} - 34i a^{11} r^2 \cos(\theta)^{11} - 66 a^{10} r^3 \cos(\theta)^{10} + 55i a^9 r^4 \cos(\theta)^9 - 33 a^8 r^5 \cos(\theta)^8 + 132i a^7 r^6 \cos(\theta)^7 + 132 a^6 r^7 \cos(\theta)^6 - 33i a^5 r^8 \cos(\theta)^5 + 55 a^4 r^9 \cos(\theta)^4 - 33 a^3 r^{10} \cos(\theta)^3 + 5 a^2 r^{11} \cos(\theta)^2 - a r^{12} \cos(\theta) + r^{13} \right)} \\
& + \frac{5 a^{10} r^2 \cos(\theta)^{10}}{i a^{13} \cos(\theta)^{13} + 9 a^{12} r \cos(\theta)^{12} - 34i a^{11} r^2 \cos(\theta)^{11} - 66 a^{10} r^3 \cos(\theta)^{10} + 55i a^9 r^4 \cos(\theta)^9 - 33 a^8 r^5 \cos(\theta)^8 + 132i a^7 r^6 \cos(\theta)^7 + 132 a^6 r^7 \cos(\theta)^6 - 33i a^5 r^8 \cos(\theta)^5 + 55 a^4 r^9 \cos(\theta)^4 - 33 a^3 r^{10} \cos(\theta)^3 + 5 a^2 r^{11} \cos(\theta)^2 - a r^{12} \cos(\theta) + r^{13}} \\
& - \dots
\end{aligned}$$

Relation between Spin Coefficients

$$\begin{aligned}
\bar{\tau} + \pi = & \frac{i\sqrt{2}a^9 \cos(\theta)^8 \sin(\theta)}{2\left(a^{10} \cos(\theta)^{10} + 8i a^9 r \cos(\theta)^9 - 27 a^8 r^2 \cos(\theta)^8 - 48i a^7 r^3 \cos(\theta)^7 + 42 a^6 r^4 \cos(\theta)^6 + 42 a^4 r^6 \cos(\theta)^4 + 48i a^3 r^7 \cos(\theta)^3 - 27 a^2 r^8 \cos(\theta)^2 - 8i a r^9 \cos(\theta) + r^{10}\right)} \\
& - \frac{4\sqrt{2}a^8 r \cos(\theta)^7 \sin(\theta)}{a^{10} \cos(\theta)^{10} + 8i a^9 r \cos(\theta)^9 - 27 a^8 r^2 \cos(\theta)^8 - 48i a^7 r^3 \cos(\theta)^7 + 42 a^6 r^4 \cos(\theta)^6 + 42 a^4 r^6 \cos(\theta)^4 + 48i a^3 r^7 \cos(\theta)^3 - 27 a^2 r^8 \cos(\theta)^2 - 8i a r^9 \cos(\theta) + r^{10}} \\
& - \frac{14i\sqrt{2}a^7 r^2 \cos(\theta)^6 \sin(\theta)}{a^{10} \cos(\theta)^{10} + 8i a^9 r \cos(\theta)^9 - 27 a^8 r^2 \cos(\theta)^8 - 48i a^7 r^3 \cos(\theta)^7 + 42 a^6 r^4 \cos(\theta)^6 + 42 a^4 r^6 \cos(\theta)^4 + 48i a^3 r^7 \cos(\theta)^3 - 27 a^2 r^8 \cos(\theta)^2 - 8i a r^9 \cos(\theta) + r^{10}} \\
& + \frac{28\sqrt{2}a^6 r^3 \cos(\theta)^5 \sin(\theta)}{a^{10} \cos(\theta)^{10} + 8i a^9 r \cos(\theta)^9 - 27 a^8 r^2 \cos(\theta)^8 - 48i a^7 r^3 \cos(\theta)^7 + 42 a^6 r^4 \cos(\theta)^6 + 42 a^4 r^6 \cos(\theta)^4 + 48i a^3 r^7 \cos(\theta)^3 - 27 a^2 r^8 \cos(\theta)^2 - 8i a r^9 \cos(\theta) + r^{10}} \\
& + \frac{35i\sqrt{2}a^5 r^4 \cos(\theta)^4 \sin(\theta)}{a^{10} \cos(\theta)^{10} + 8i a^9 r \cos(\theta)^9 - 27 a^8 r^2 \cos(\theta)^8 - 48i a^7 r^3 \cos(\theta)^7 + 42 a^6 r^4 \cos(\theta)^6 + 42 a^4 r^6 \cos(\theta)^4 + 48i a^3 r^7 \cos(\theta)^3 - 27 a^2 r^8 \cos(\theta)^2 - 8i a r^9 \cos(\theta) + r^{10}} \\
& - \frac{i\sqrt{2}a^7 \cos(\theta)^6 \sin(\theta)}{2\left(a^8 \cos(\theta)^8 + 8i a^7 r \cos(\theta)^7 - 28 a^6 r^2 \cos(\theta)^6 - 56i a^5 r^3 \cos(\theta)^5 + 70 a^4 r^4 \cos(\theta)^4 + 56i a^3 r^5 \cos(\theta)^3 - 28 a^2 r^6 \cos(\theta)^2 - 8i a r^7 \cos(\theta) + r^8\right)} \\
& - \frac{28\sqrt{2}a^4 r^5 \cos(\theta)^3 \sin(\theta)}{a^{10} \cos(\theta)^{10} + 8i a^9 r \cos(\theta)^9 - 27 a^8 r^2 \cos(\theta)^8 - 48i a^7 r^3 \cos(\theta)^7 + 42 a^6 r^4 \cos(\theta)^6 + 42 a^4 r^6 \cos(\theta)^4 + 48i a^3 r^7 \cos(\theta)^3 - 27 a^2 r^8 \cos(\theta)^2 - 8i a r^9 \cos(\theta) + r^{10}} \\
& + \frac{3\sqrt{2}a^6 r \cos(\theta)^5 \sin(\theta)}{a^8 \cos(\theta)^8 + 8i a^7 r \cos(\theta)^7 - 28 a^6 r^2 \cos(\theta)^6 - 56i a^5 r^3 \cos(\theta)^5 + 70 a^4 r^4 \cos(\theta)^4 + 56i a^3 r^5 \cos(\theta)^3 - 28 a^2 r^6 \cos(\theta)^2 - 8i a r^7 \cos(\theta) + r^8} \\
& - \frac{14i\sqrt{2}a^3 r^6 \cos(\theta)^2 \sin(\theta)}{a^{10} \cos(\theta)^{10} + 8i a^9 r \cos(\theta)^9 - 27 a^8 r^2 \cos(\theta)^8 - 48i a^7 r^3 \cos(\theta)^7 + 42 a^6 r^4 \cos(\theta)^6 + 42 a^4 r^6 \cos(\theta)^4 + 48i a^3 r^7 \cos(\theta)^3 - 27 a^2 r^8 \cos(\theta)^2 - 8i a r^9 \cos(\theta) + r^{10}} + \dots
\end{aligned}$$

$$\bar{\alpha} + \beta$$

$$\begin{aligned}
&= \frac{a^8 \cos(\theta)^9}{2 \left(i \sqrt{2} a^9 \cos(\theta)^9 + 9 \sqrt{2} a^8 r \cos(\theta)^8 - 36 i \sqrt{2} a^7 r^2 \cos(\theta)^7 - 84 \sqrt{2} a^6 r^3 \cos(\theta)^6 + 126 i \sqrt{2} a^5 r^4 \cos(\theta)^5 + 126 \sqrt{2} a^4 r^5 \cos(\theta)^4 - 84 i \sqrt{2} a^3 r^6 \cos(\theta)^3 - 36 \sqrt{2} a^2 r^7 \cos(\theta)^2 + 9 i \sqrt{2} a r^8 \cos(\theta) - r^9 \right)} \\
&\quad - \frac{3 i a^7 r \cos(\theta)^8}{\left(i \sqrt{2} a^9 \cos(\theta)^9 + 9 \sqrt{2} a^8 r \cos(\theta)^8 - 36 i \sqrt{2} a^7 r^2 \cos(\theta)^7 - 84 \sqrt{2} a^6 r^3 \cos(\theta)^6 + 126 i \sqrt{2} a^5 r^4 \cos(\theta)^5 + 126 \sqrt{2} a^4 r^5 \cos(\theta)^4 - 84 i \sqrt{2} a^3 r^6 \cos(\theta)^3 - 36 \sqrt{2} a^2 r^7 \cos(\theta)^2 + 9 i \sqrt{2} a r^8 \cos(\theta) - r^9 \right)} \\
&\quad - \frac{a^8 \cos(\theta)^7}{\left(i \sqrt{2} a^9 \cos(\theta)^9 + 9 \sqrt{2} a^8 r \cos(\theta)^8 - 36 i \sqrt{2} a^7 r^2 \cos(\theta)^7 - 84 \sqrt{2} a^6 r^3 \cos(\theta)^6 + 126 i \sqrt{2} a^5 r^4 \cos(\theta)^5 + 126 \sqrt{2} a^4 r^5 \cos(\theta)^4 - 84 i \sqrt{2} a^3 r^6 \cos(\theta)^3 - 36 \sqrt{2} a^2 r^7 \cos(\theta)^2 + 9 i \sqrt{2} a r^8 \cos(\theta) - r^9 \right)} \\
&\quad - \frac{7 a^6 r^2 \cos(\theta)^7}{\left(i \sqrt{2} a^9 \cos(\theta)^9 + 9 \sqrt{2} a^8 r \cos(\theta)^8 - 36 i \sqrt{2} a^7 r^2 \cos(\theta)^7 - 84 \sqrt{2} a^6 r^3 \cos(\theta)^6 + 126 i \sqrt{2} a^5 r^4 \cos(\theta)^5 + 126 \sqrt{2} a^4 r^5 \cos(\theta)^4 - 84 i \sqrt{2} a^3 r^6 \cos(\theta)^3 - 36 \sqrt{2} a^2 r^7 \cos(\theta)^2 + 9 i \sqrt{2} a r^8 \cos(\theta) - r^9 \right)} \\
&\quad + \frac{7 i a^7 r \cos(\theta)^6}{\left(i \sqrt{2} a^9 \cos(\theta)^9 + 9 \sqrt{2} a^8 r \cos(\theta)^8 - 36 i \sqrt{2} a^7 r^2 \cos(\theta)^7 - 84 \sqrt{2} a^6 r^3 \cos(\theta)^6 + 126 i \sqrt{2} a^5 r^4 \cos(\theta)^5 + 126 \sqrt{2} a^4 r^5 \cos(\theta)^4 - 84 i \sqrt{2} a^3 r^6 \cos(\theta)^3 - 36 \sqrt{2} a^2 r^7 \cos(\theta)^2 + 9 i \sqrt{2} a r^8 \cos(\theta) - r^9 \right)} \\
&\quad + \frac{7 i a^5 r^3 \cos(\theta)^6}{\left(i \sqrt{2} a^9 \cos(\theta)^9 + 9 \sqrt{2} a^8 r \cos(\theta)^8 - 36 i \sqrt{2} a^7 r^2 \cos(\theta)^7 - 84 \sqrt{2} a^6 r^3 \cos(\theta)^6 + 126 i \sqrt{2} a^5 r^4 \cos(\theta)^5 + 126 \sqrt{2} a^4 r^5 \cos(\theta)^4 - 84 i \sqrt{2} a^3 r^6 \cos(\theta)^3 - 36 \sqrt{2} a^2 r^7 \cos(\theta)^2 + 9 i \sqrt{2} a r^8 \cos(\theta) - r^9 \right)} \\
&\quad + \frac{21 a^6 r^2 \cos(\theta)^5}{\left(i \sqrt{2} a^9 \cos(\theta)^9 + 9 \sqrt{2} a^8 r \cos(\theta)^8 - 36 i \sqrt{2} a^7 r^2 \cos(\theta)^7 - 84 \sqrt{2} a^6 r^3 \cos(\theta)^6 + 126 i \sqrt{2} a^5 r^4 \cos(\theta)^5 + 126 \sqrt{2} a^4 r^5 \cos(\theta)^4 - 84 i \sqrt{2} a^3 r^6 \cos(\theta)^3 - 36 \sqrt{2} a^2 r^7 \cos(\theta)^2 + 9 i \sqrt{2} a r^8 \cos(\theta) - r^9 \right)} \\
&\quad - \dots
\end{aligned}$$

Ricci Tensor Components

$$\Phi_{00} = 0$$

$$\Phi_{01} = 0$$

$$\Phi_{02} = 0$$

$$\Phi_{10} = 0$$

$$\Phi_{11} = \frac{Q^2}{2 \left(a^4 \cos(\theta)^4 + 2 a^2 r^2 \cos(\theta)^2 + r^4 \right)}$$

$$\Phi_{12} = 0$$

$$\Phi_{20} = 0$$

$$\Phi_{21} = 0$$

$$\Phi_{22} = 0$$

$$\Lambda = 0$$

Weyl Tensor Components

$$\Psi_0 = 0$$

$$\Psi_1 = 0$$

$$\begin{aligned} \Psi_2 = & \frac{i M a^3 \cos(\theta)^3}{a^6 \cos(\theta)^6 + 3 a^4 r^2 \cos(\theta)^4 + 3 a^2 r^4 \cos(\theta)^2 + r^6} - \frac{Q^2 a^2 \cos(\theta)^2}{a^6 \cos(\theta)^6 + 3 a^4 r^2 \cos(\theta)^4 + 3 a^2 r^4 \cos(\theta)^2 + r^6} + \frac{3 M a^2 r \cos(\theta)^2}{a^6 \cos(\theta)^6 + 3 a^4 r^2 \cos(\theta)^4 + 3 a^2 r^4 \cos(\theta)^2 + r^6} \\ & + \frac{2i Q^2 a r \cos(\theta)}{a^6 \cos(\theta)^6 + 3 a^4 r^2 \cos(\theta)^4 + 3 a^2 r^4 \cos(\theta)^2 + r^6} - \frac{3i M a r^2 \cos(\theta)}{a^6 \cos(\theta)^6 + 3 a^4 r^2 \cos(\theta)^4 + 3 a^2 r^4 \cos(\theta)^2 + r^6} \\ & + \frac{Q^2 r^2}{a^6 \cos(\theta)^6 + 3 a^4 r^2 \cos(\theta)^4 + 3 a^2 r^4 \cos(\theta)^2 + r^6} - \frac{M r^3}{a^6 \cos(\theta)^6 + 3 a^4 r^2 \cos(\theta)^4 + 3 a^2 r^4 \cos(\theta)^2 + r^6} \end{aligned}$$

$$\Psi_3 = 0$$

$$\Psi_4 = 0$$

Newman-Penrose Equations

Equations are vanished.

Bianchi Identities

Equations are vanished.

Petrov Type

Type " D "